Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 3 Probability & Statistics

1. A factory has an automatic machine that produces an important part. Past records indicate that at the beginning of the day the machine is set up correctly 70 percent of the time. Past experience also shows that if the machine is set up correctly it will produce good parts 90 percent of the time. If it is set up incorrectly, it will still produce good parts 40 percent of the time. If the machine is set up and produces a good part, what is the probability that it is set up correctly?

2. Let X be the number of years a particular type of machine is in good working condition and need no replacement. Assume that X has the probability mass function

$$P(1) = 0.4, \quad P(2) = 0.3, \quad P(3) = 0.2, \quad P(4) = 0.08, \quad P(5) = 0.02.$$

   Find the distribution function F. Sketch F.

   What is the probability that the machine needs no replacement during the first 3 years.

3. A supplier of kerosene begins each week with a supply of 200 $k\ell$ kerozene. He notices that his weekly demands show a relative frequency behavior that increases steadily up to 100 $k\ell$, and then levels off between 100 and 200 $k\ell$. Let X denote weekly demand in hundreds of $k\ell$. Then, the relative frequencies for demand are adequately modeled by

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \le x \le 1 \\ \frac{1}{2}, & 1 < x \le 2 \\ 0, & x > 2. \end{cases}$$

   Find the distribution function F. Sketch F.

Answers

1. 0.84

2. $F(x) = \begin{cases} 0, & x < 1 \\ 0.4, & 1 \le x < 2 \\ 0.7, & 2 \le x < 3 \\ 0.9, & 3 \le x < 4 \\ 0.98, & 4 \le x < 5 \\ 1, & x \ge 5 \end{cases}$

   prob $= 0.3$

3. $F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}\, x^2, & 0 \le x < 1 \\ \frac{1}{2}\, x, & 1 \le x < 2 \\ 1, & x \ge 2 \end{cases}$